
FIREWALL CREATION AND TESTING

INTRODUCTION

A firewall is a critical security mechanism designed to monitor and control network traffic based on predefined rules. It acts as a barrier between trusted internal networks and untrusted external networks, ensuring that only authorised communication is allowed. In configuring a firewall, administrators define profiles and rules that determine how traffic flows in and out of the system.

NETWORK PROFILE

- **Domain Profile:** Applied when the computer is connected to a domain network (e.g., an organization's Active Directory). It assumes a higher level of trust and allows more communication for business operations.
- **Private Profile:** Used for trusted networks such as home or small office setups. It permits sharing and discovery features but still enforces protection against external threats.
- **Public Profile:** Activated when connected to public or untrusted networks (e.g., airports, cafés). This is the most restrictive profile, blocking most inbound connections to minimize exposure.

TRAFFIC DIRECTION RULES

Firewalls also distinguish traffic based on its direction:

- **Inbound Rules:** Control traffic coming **into** the system from external sources. For example, blocking unsolicited connections from the internet or allowing specific services like SSH or HTTP.
- **Outbound Rules:** Govern traffic **leaving** the system. These rules can restrict applications from sending data out, preventing unauthorized communication or data exfiltration.

OBJECTIVE

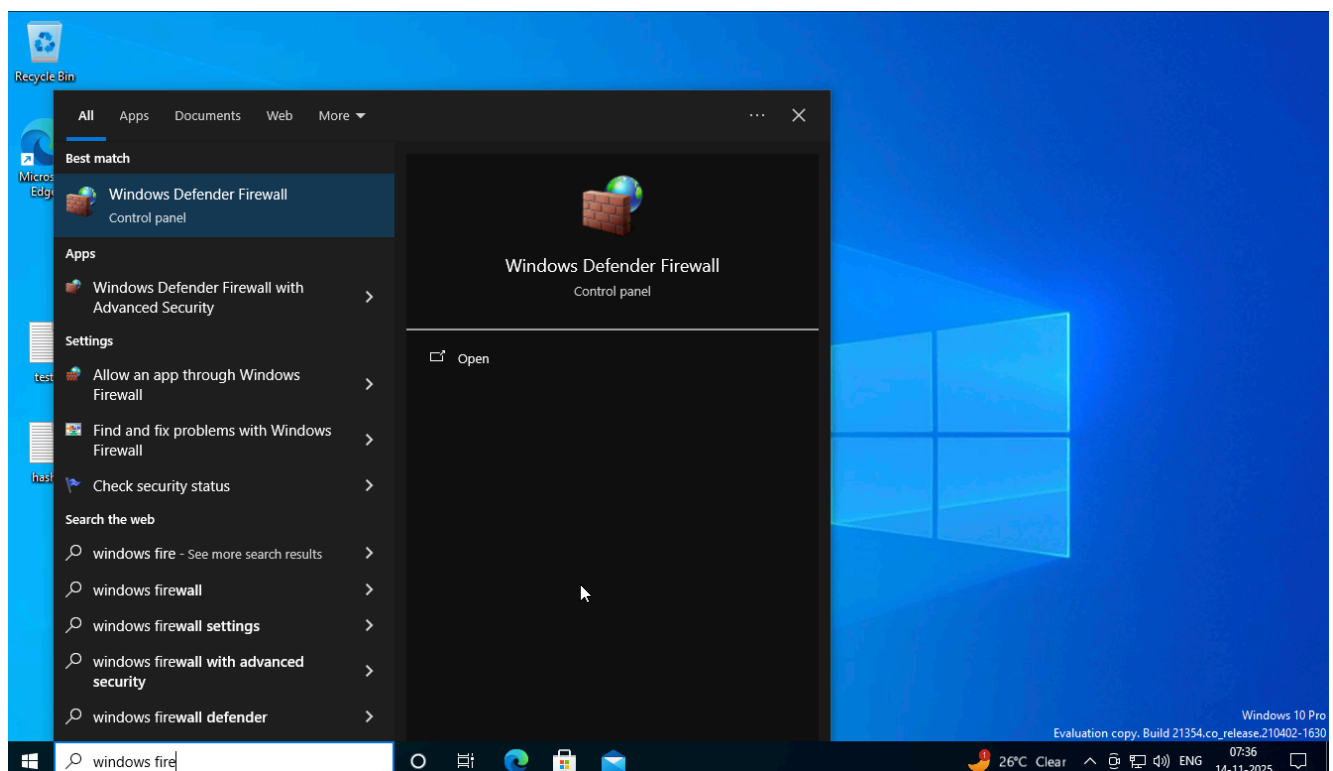
To block outbound HTTP traffic (TCP port 80) using Windows Firewall and verifying the rule with before and after tests.

RULE

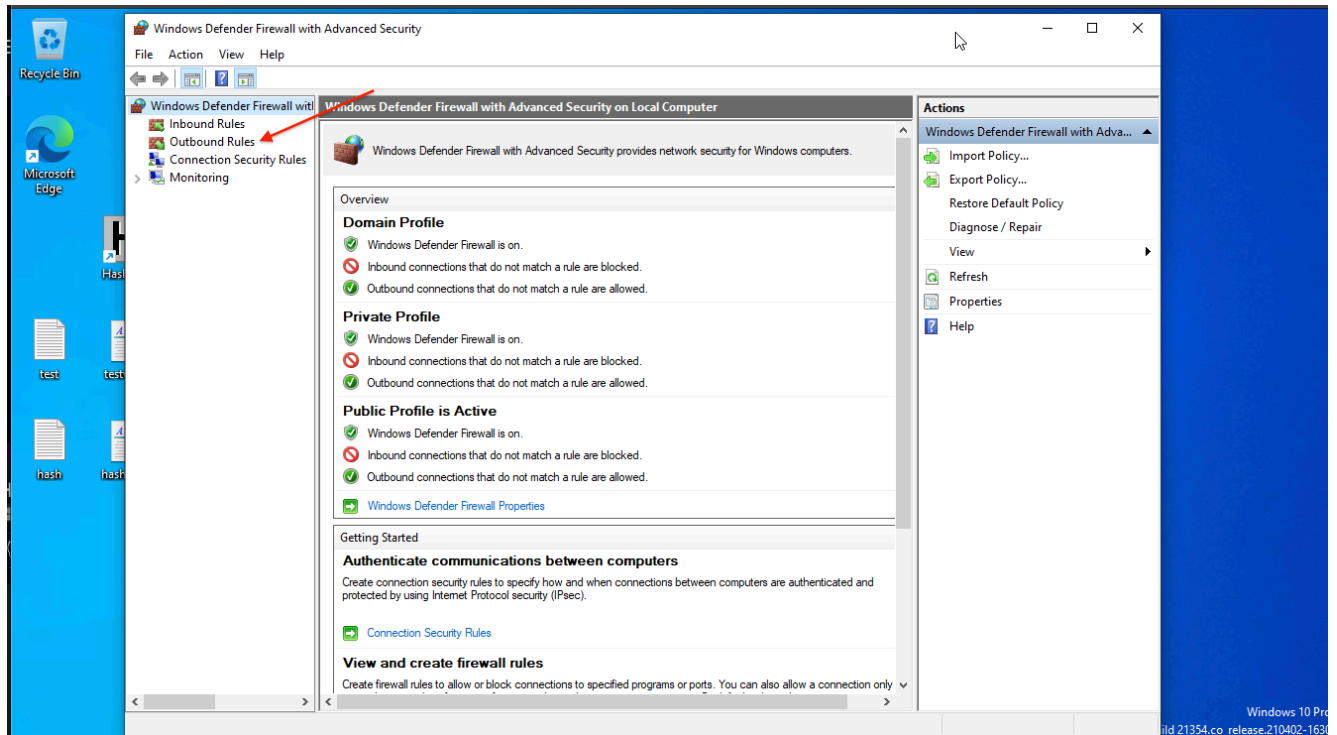
- **Rule Name:** Block HTTP Outbound
- **Direction:** Outbound
- **Protocol:** TCP
- **Port:** 80
- **Action:** Block connection
- **Scope:** All profiles (Domain, Private, Public)

STEPS TO CREATE THE RULE

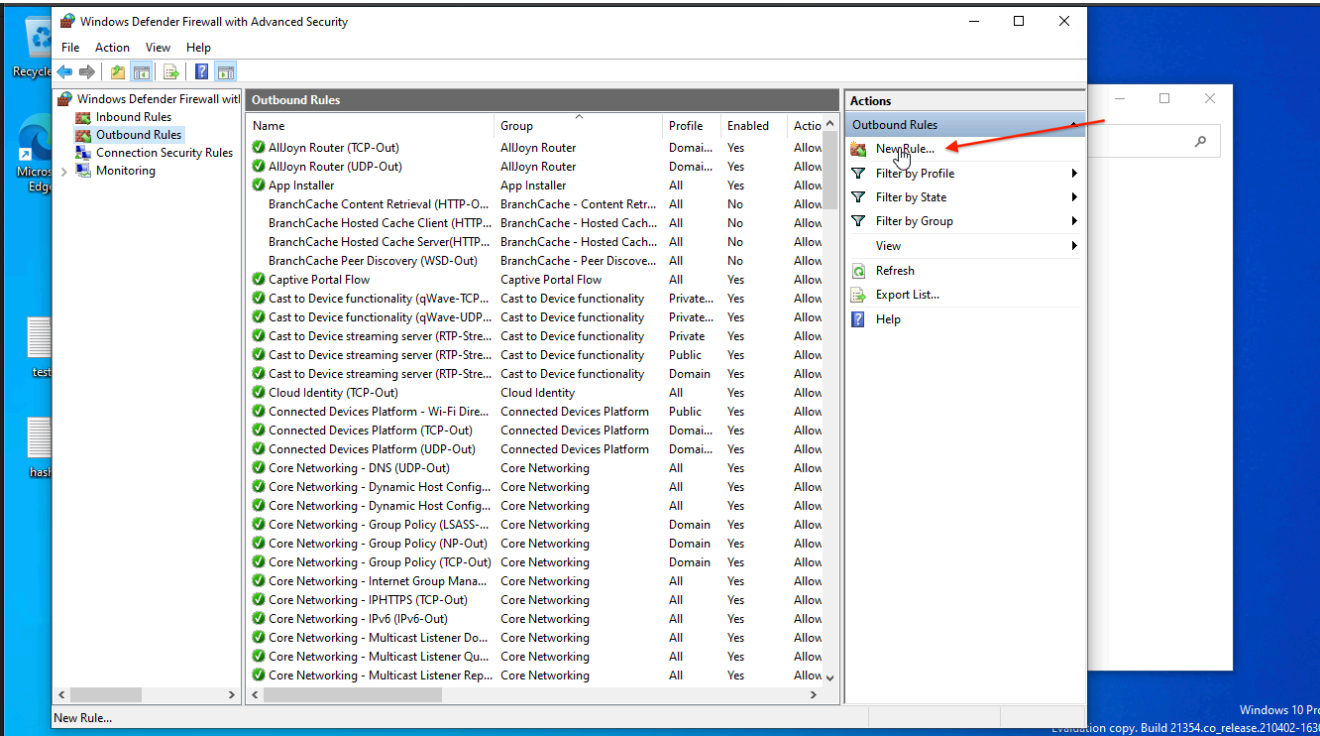
STEP 1 : Press win + R or type Windows Defender in the windows search panel and open Windows Defender Firewall.



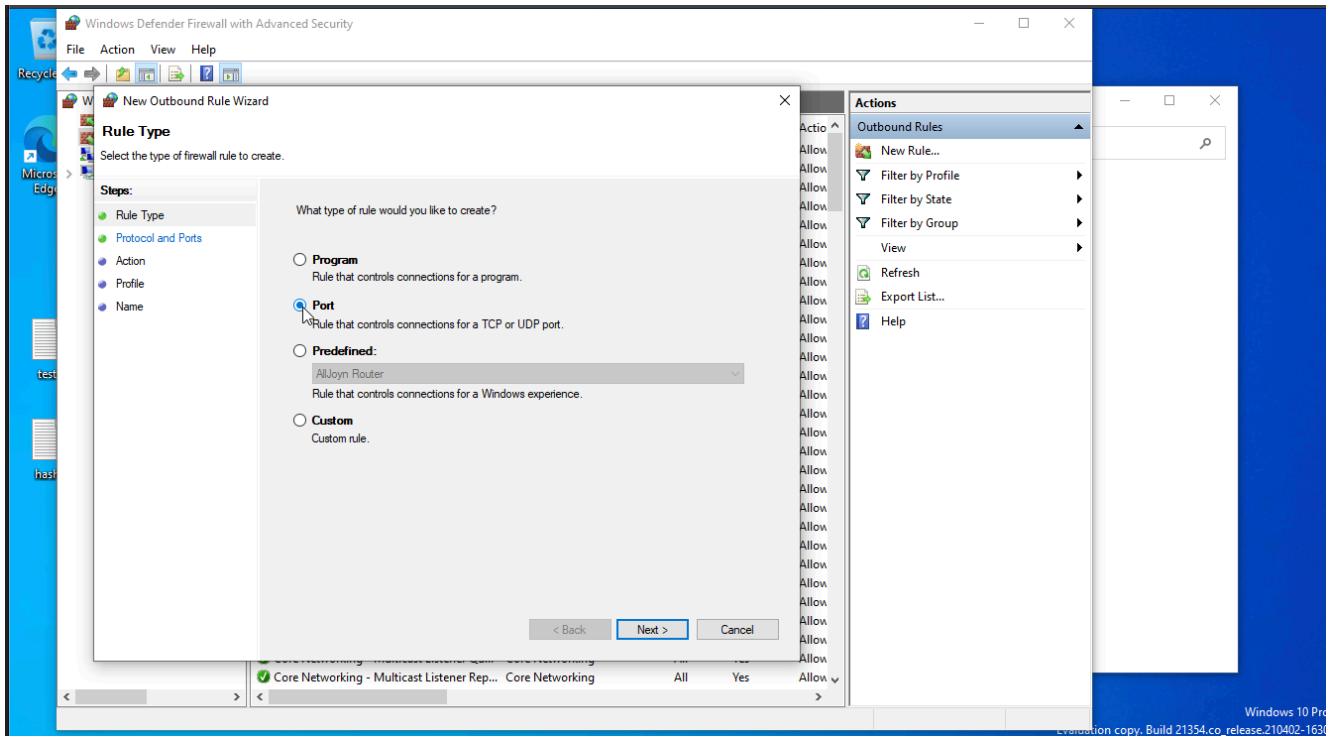
STEP 2 : In the left panel under Windows Defender Firewall with Advanced Security, click on “Outbound rules”



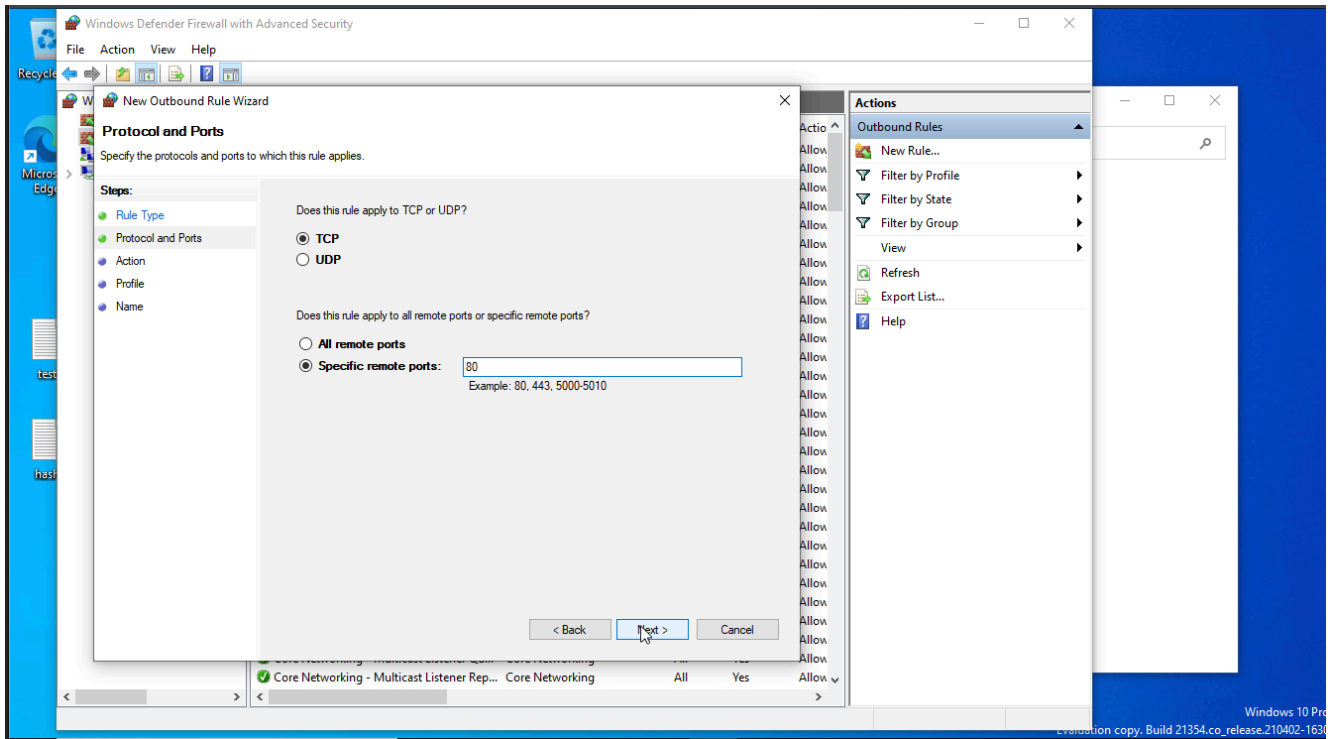
STEP 3 : In the right panel click on “New Rule”.



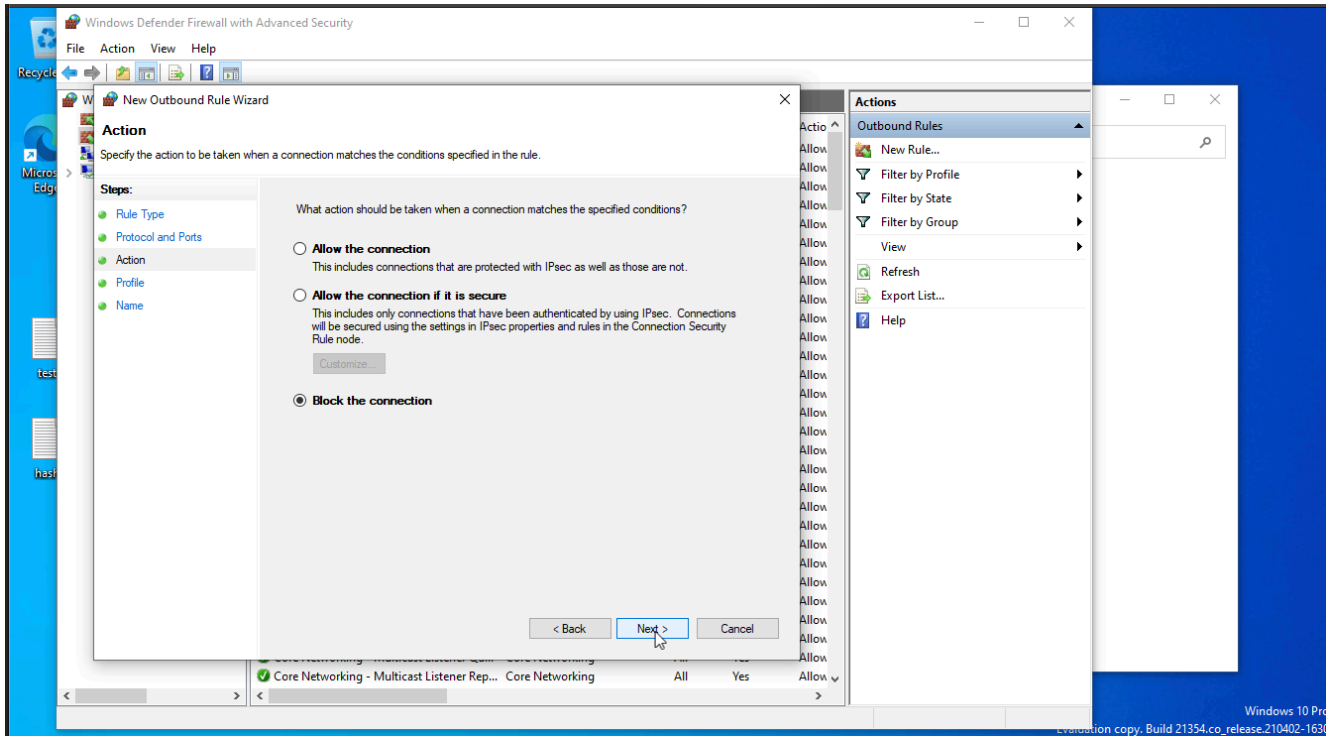
STEP 4 : Under the Rule type section, select “Port” and click on “Next”.



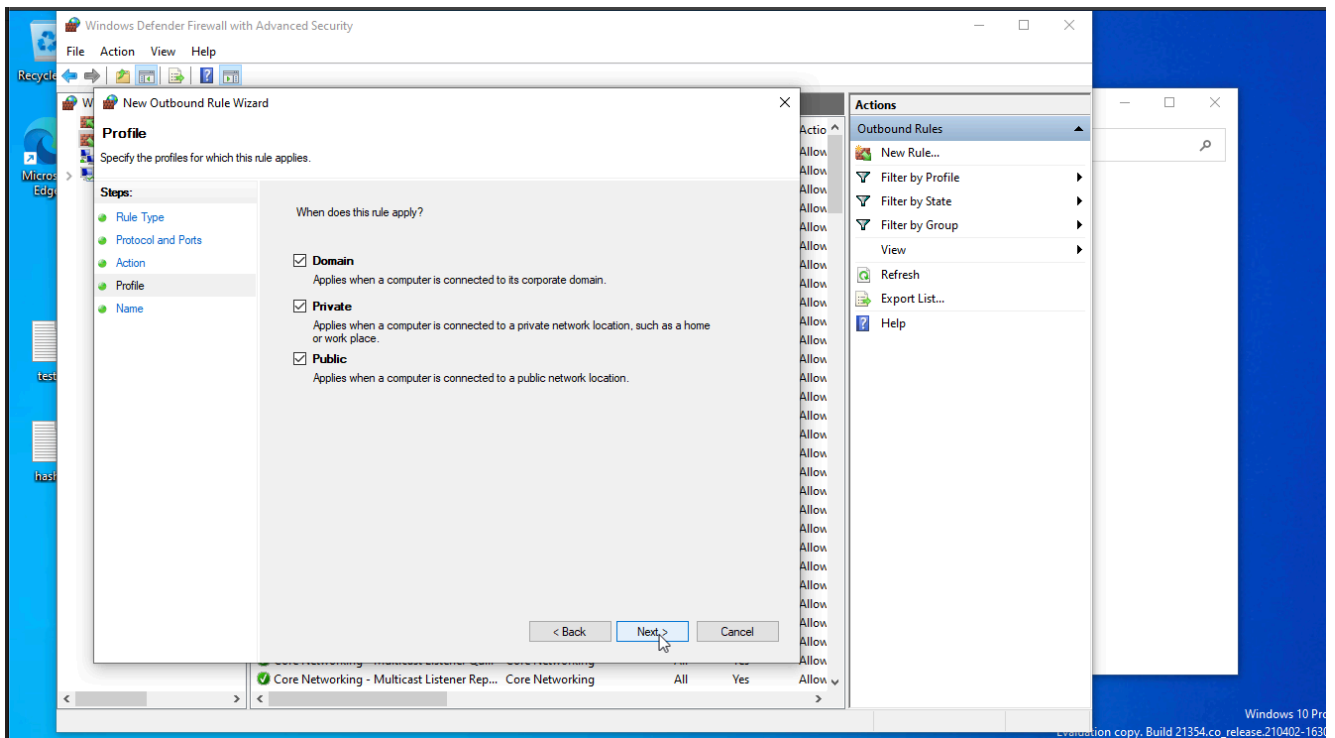
STEP 5 : In the Protocols and Port section, select TCP and enter 80 in the Specific remote ports, since HTTP is on port 80.



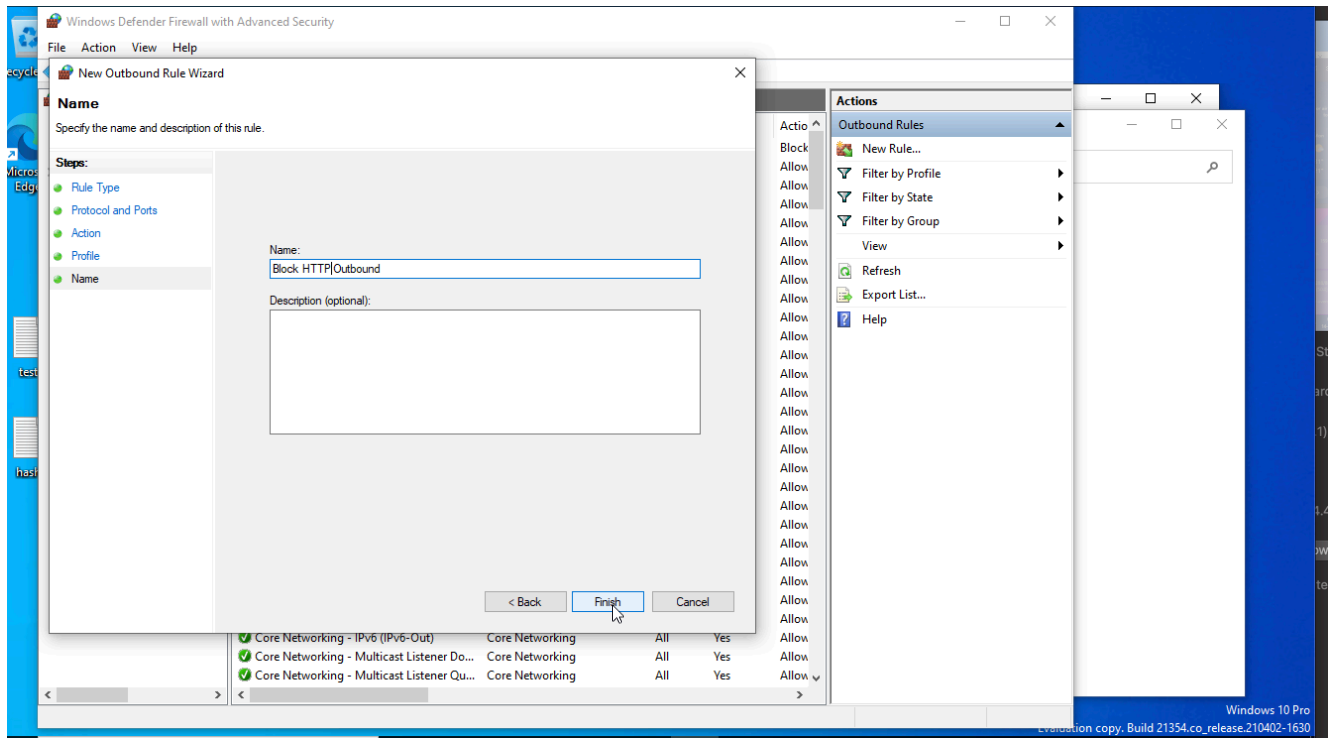
STEP 6 : In the Action section select “Block the connection” and click on “Next”.



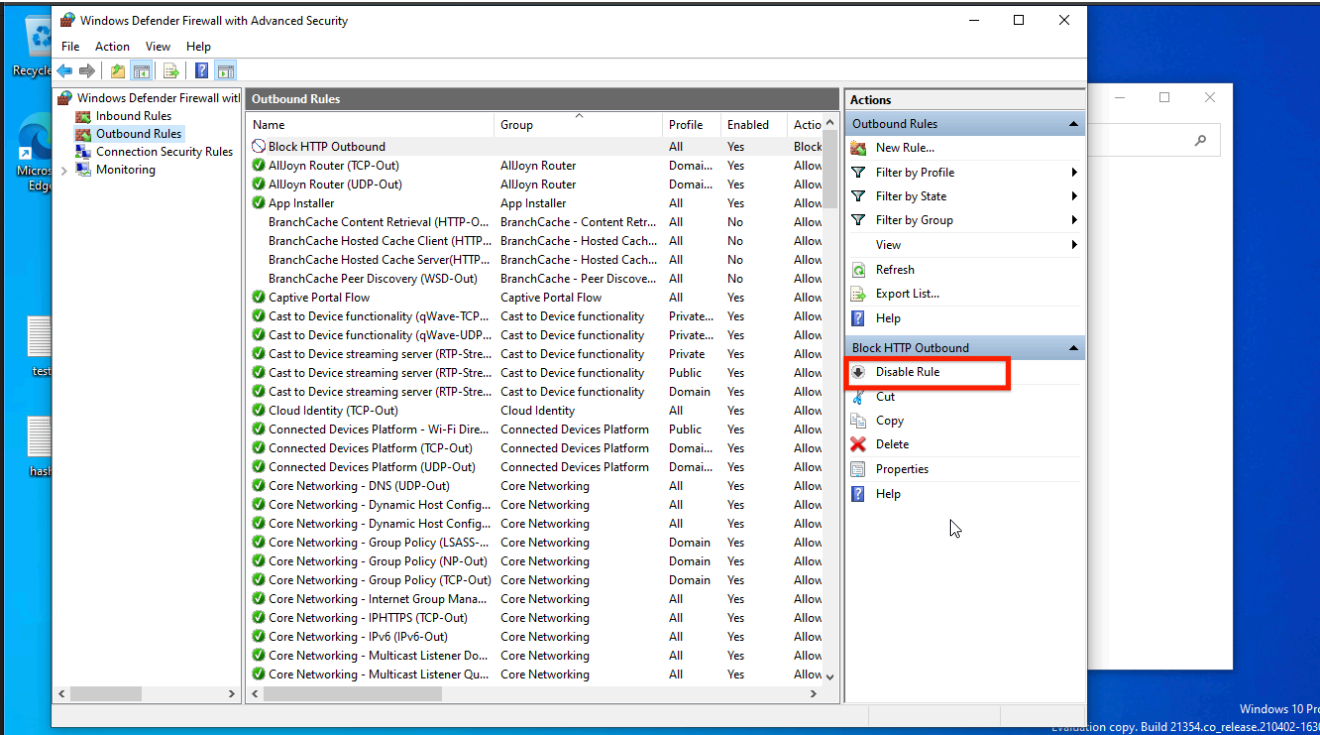
STEP 7 : In the Profile section, enable all the three options i.e Domain, Private and Public to ensure that the rule is applied on any types of network.



STEP 8 : In the name section, give the name to the firewall in accordance with its function. Here I have given “Block HTTP Outbound”. Then click on “Finish” this will create the rule.

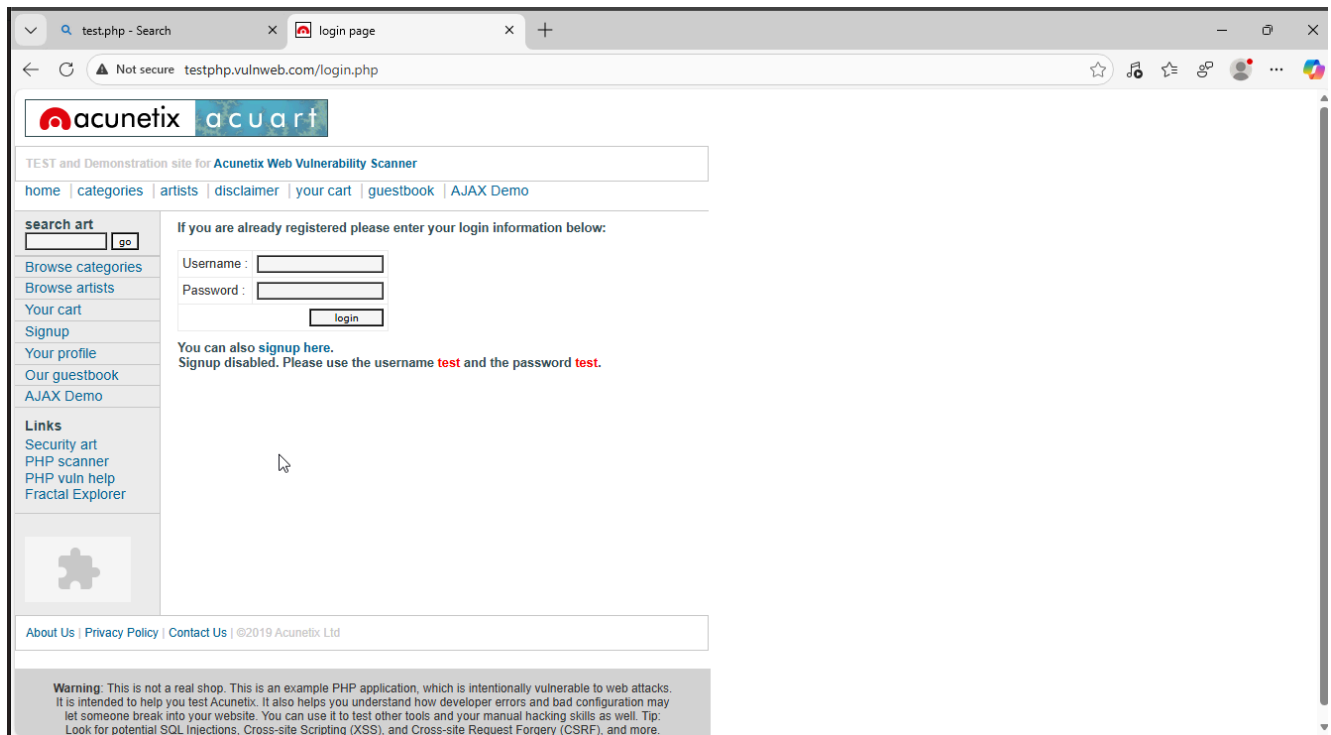


STEP 9 : Make sure that the rule is enabled.

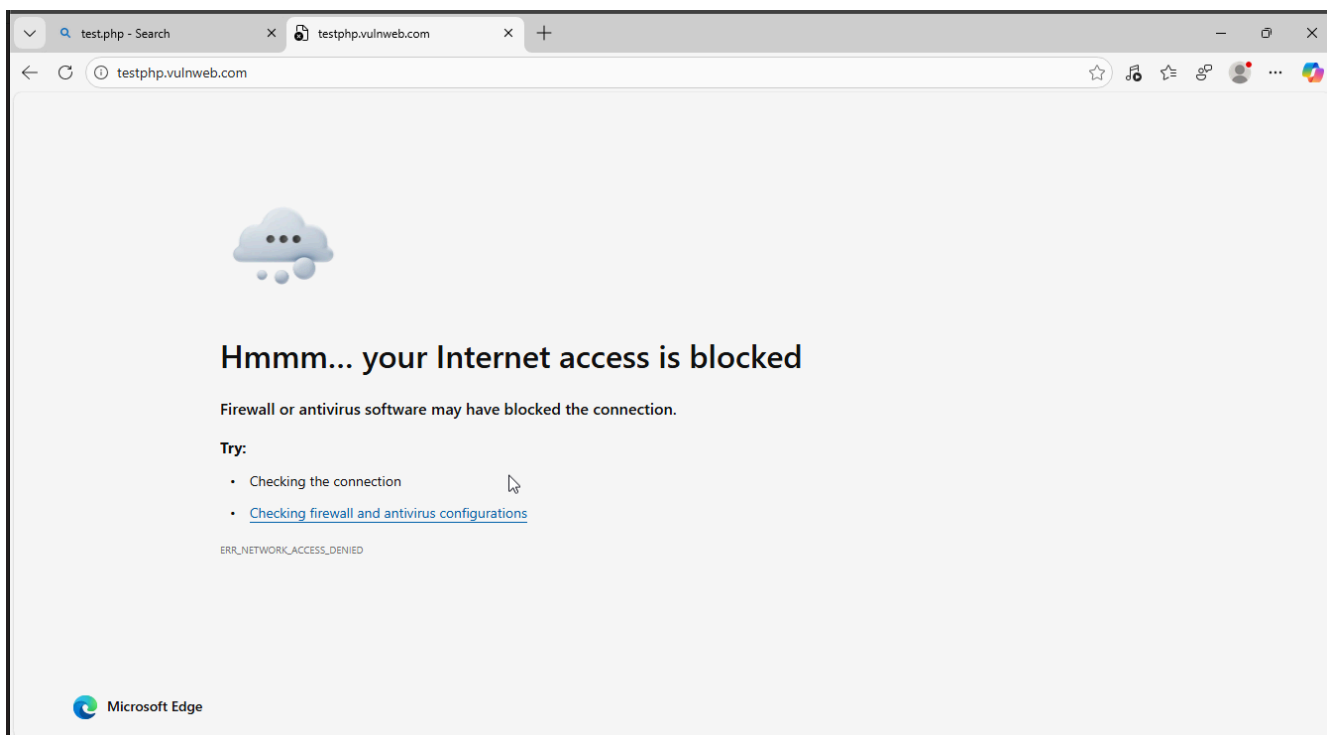


STEP 10 : Verify the rule.

A) BEFORE : The computer could send and receive traffic from HTTP server.



B) AFTER : The computer blocked the HTTP traffic and the connection failed.



CONCLUSION

The firewall rule successfully blocked outbound HTTP traffic on port 80. This shows how Windows Firewall can be used to control network communication and enforce security policies